Mini-project (Course 2)

Applying supervised learning to predict student dropout.

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Date: 14-Mar-2026

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1.0 Overview: Problem Statement

The project requires examination of student data (Data, 2026) to develop a predictive model on student dropout. Detecting dropouts early lead to earlier interventions preventing dropouts. A high dropout rate means high revenue loss, drop in reputation and student satisfaction.

2.0 Approach

Approach (Figure 2.1) taken with two supervised models – XGBoost and Neural Networks, including hyperparameter tuning

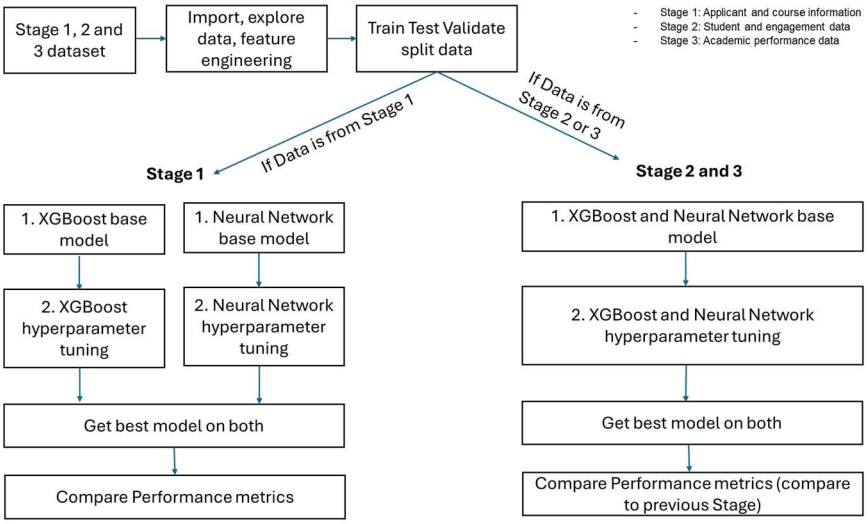


Figure 2.1: Approach workflow

2.1 EDA and Feature Engineering

16 features with 25,059 rows (Figure 2.2).

```
data.head()
```

```
(25059, 16)
```

	CentreName	LearnerCode	BookingType	LeadSource	DiscountType	DateofBirth	Gender	Nationality	HomeState	HomeCity	Coursel
0	ISC_Aberdeen	2284932	Agent	Standard Agent Booking	NaN	13/01/1998	Male	Chinese	Jianye District; Jiangsu Province	Nanjing	Pre-Ma
1	ISC_Aberdeen	2399500	Agent	Standard Agent Booking	NaN	12/2/1998	Male	Chinese	NaN	Xi'an	Founc
2	ISC_Aberdeen	2424946	Agent	Standard Agent Booking	NaN	7/4/2000	Male	Chinese	NaN	Chengdu	Interna Yea
3	ISC_Aberdeen	2426583	Agent	Standard Agent Booking	NaN	18/05/1999	Male	Chinese	NaN	LIUYANG CITY ;CHANGSHA	Interna Yea
4	ISC_Aberdeen	2434674	Agent	Standard Agent Booking	NaN	19/04/1996	Male	Chinese	NaN	Xiamen	Pre-Ma

Figure 2.2: Top 5 data rows

The ‘LearnerCode’ (student id) removed, 4 features > 200 unique values, DateOfBirth converted to Age. Other 3 features (figure 2.3) removed. CourseLevel - ordinal encoder performed. One-hot encoding performed on remaining categorical data.

```
# Show metadata of data.
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25059 entries, 0 to 25058
Data columns (total 15 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   CentreName             25059 non-null  object
 1   BookingType            25059 non-null  object
 2   LeadSource             25059 non-null  object
 3   DiscountType           7595 non-null   object
 4   DateofBirth            25059 non-null  object
 5   Gender                 25059 non-null  object
 6   Nationality            25059 non-null  object
 7   HomeState              8925 non-null   object
 8   HomeCity               21611 non-null  object
 9   CourseLevel            25059 non-null  object
 10  CourseName             25059 non-null  object
 11  IsFirstIntake          25059 non-null  bool
 12  CompletedCourse        25059 non-null  object
 13  ProgressionDegree      24184 non-null  object
 14  ProgressionUniversity  25059 non-null  object
dtypes: bool(1), object(14)
memory usage: 2.7+ MB
```

Above shows type of data

```
count of unique values in column

CentreName                19
BookingType                2
LeadSource                 7
DiscountType              11
DateofBirth             4705
Gender                     2
Nationality              151
HomeState               2448
HomeCity                5881
CourseLevel                4
CourseName               172
IsFirstIntake              2
CompletedCourse            2
ProgressionDegree       2616
ProgressionUniversity     40
dtype: int64
```

Columns with > 200 unique values

```
unique values in column

CentreName                [ISC_Aberdeen, ISC_Cardiff, ISC_Dublin, ISC_Du...
BookingType                                                [Agent, Direct]
LeadSource                [Standard Agent Booking, Sponsor, APD, Digital...
DiscountType              [nan, Other Scholarships / Bursaries, Bursary,...
DateofBirth               [13/01/1998, 12/2/1998, 7/4/2000, 18/05/1999, ...
Gender                                                   [Male, Female]
Nationality               [Chinese, Iraqi, Kenyan, British, Kazakhstani,...
HomeState                 [Jianye District; Jiangsu Province, nan, Hunan...
HomeCity                  [Nanjing, Xi'an, Chengdu, LIUYANG CITY ;CHANGS...
CourseLevel               [Pre-Masters, Foundation, International Year T...
CourseName                [Business and Law Pre-Masters, Life Sciences U...
IsFirstIntake                                            [True, False]
CompletedCourse                                              [Yes, No]
ProgressionDegree         [Msc Econ Accounting and Investment Management...
ProgressionUniversity     [University of Aberdeen, Cardiff University, U...
dtype: object
```

Unique values in columns. Note CourseLevel seems to be ordinal data

Figure 2.3: Stage1 data

There are > 50% null values on DiscountType feature (figure 2.4) - removed.

```
# Find missing values.
data.isnull().sum()
```

	0
CentreName	0
BookingType	0
LeadSource	0
DiscountType	17464
DateofBirth	0
Gender	0
Nationality	0
CourseLevel	0
CourseName	0
IsFirstIntake	0
CompletedCourse	0
ProgressionUniversity	0

dtype: int64

17,464 rows are null in DiscountType = 70% of the total data 925,059).

Data encoding means below shows mostly 0 and 1s of min/max. Age has range of 20-63 years, with mean of 25.24.

Describe function of data

	CourseLevel	CentreName_ISC_Cardiff	CentreName_ISC_Dublin	CentreName_ISC_Durham	CentreName_ISC_Holland	CentreName_ISC_Huddersfield	CentreName_ISC_Kingston
count	25059.000000	25059.000000	25059.000000	25059.000000	25059.000000	25059.000000	25059.000000
mean	0.887905	0.027415	0.015922	0.102558	0.017399	0.071072	0.075502
std	1.231749	0.163293	0.125178	0.303387	0.130755	0.256951	0.264205
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
75%	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
max	3.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000

8 rows × 390 columns

```
Duplicate Rows :  CourseLevel                          13463
CentreName_ISC_Cardiff                    13463
CentreName_ISC_Dublin                     13463
CentreName_ISC_Durham                     13463
CentreName_ISC_Holland                    13463
                                          ...
ProgressionUniversity_University of Surrey    13463
ProgressionUniversity_University of Sussex    13463
ProgressionUniversity_VU Amsterdam            13463
CompletedCourse_Yes                       13463
Age                                       13463
Length: 390, dtype: int64
```

There are a number of duplicates

Figure 2.4: Null/information data

The imbalanced 'CompletedCourse' feature was converted to 'dropout' (figure 2.5).

```
Target distribution (dropout):
dropout
0    0.850194
1    0.149806
Name: proportion, dtype: float64
```

Imbalanced data – 85% students stay in the course, 15% drop out

```
Distribution count (dropout):
dropout
0    21305
1     3754
Name: count, dtype: int64
Text(0.5, 1.0, 'Target class distribution - dropout')
```

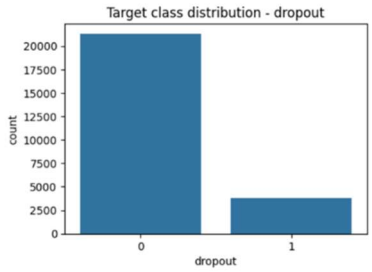


Figure 2.5: Target imbalanced distribution

Most of student in their early 20s (figure 2.6), foundation level is highest uptake.

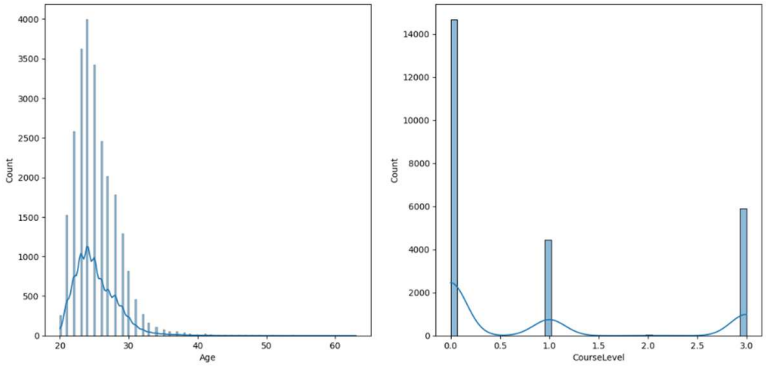


Figure 2.6: Age/CourseLevel distribution of data

The updated data (figure 2.7) was training, test split (80%-20%), validation (10% of train). Data scaled for neural networks.

```
data_df.head()
```

	CourseLevel	CentreName_ISC_Cardiff	CentreName_ISC_Dublin	CentreName_ISC_Durham	CentreName_ISC_Holland	CentreName_ISC_Huddersfield	CentreName_ISC_Kingston	CentreNa
0	3	0	0	0	0	0	0	
1	0	0	0	0	0	0	0	
2	2	0	0	0	0	0	0	
3	2	0	0	0	0	0	0	
4	3	0	0	0	0	0	0	

5 rows × 390 columns

Figure 2.7: Updated data

3.0 Stage1

3.1 XGBoost and hyperparameter tuning

XGBoost fitted on the training dataset (figure 3.1)

```
XGBoost Model Accuracy:  0.8954509177972865
XGBoost Model AUC:  0.7525924129635805
              precision    recall  f1-score   support

           0       0.92      0.96      0.94      4261
           1       0.69      0.55      0.61       751

    accuracy                           0.90      5012
   macro avg       0.81      0.75      0.78      5012
weighted avg       0.89      0.90      0.89      5012
```

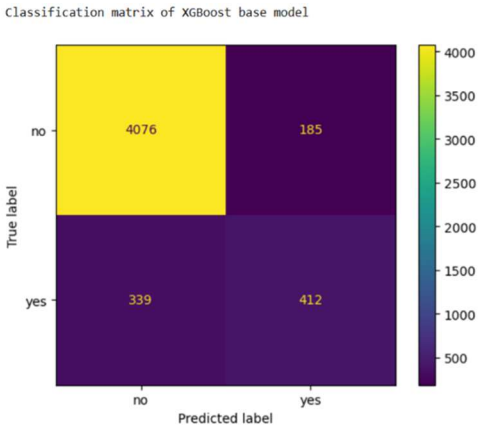


Figure 3.1: XGBoost

Test accuracy and AUC aren't high. Recall is identifying only 55% of all actual dropouts. False negatives (339) are missed opportunity.

XGBoost tuning for n_estimators (50, 100, 200), learning rate (0.05, 0.1, 0.2, 0.3), max_depth (3,5,7). AUC increased (figure3.2).

Best model

index	model	best_cv_auc	test_auc	test_accuracy	best_params
0	XGBoost Hyperparameter	0.8796735006386289	0.885334769161731	0.8936552274541102	{'learning_rate': 0.2, 'max_depth': 5, 'n_estimators': 50}

	precision	recall	f1-score	support
0	0.92	0.96	0.94	4261
1	0.69	0.53	0.60	751
accuracy			0.89	5012
macro avg	0.80	0.74	0.77	5012
weighted avg	0.89	0.89	0.89	5012

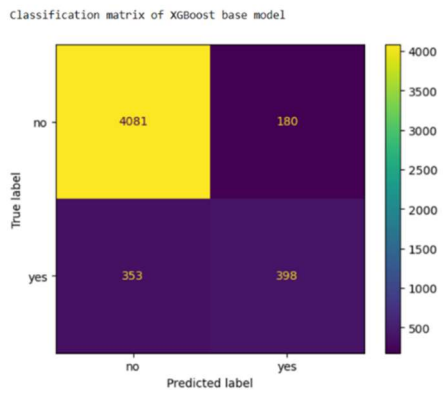


Figure 3.2: XGBoost tuned

Feature importance (figure 3.3) has nationality at top

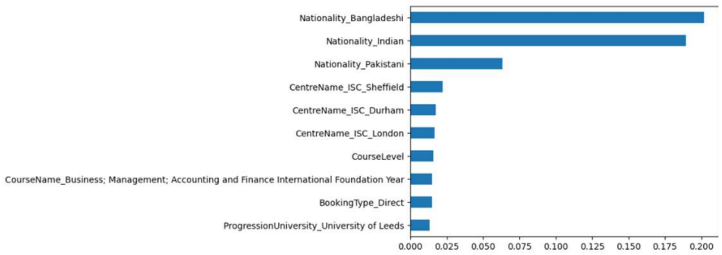
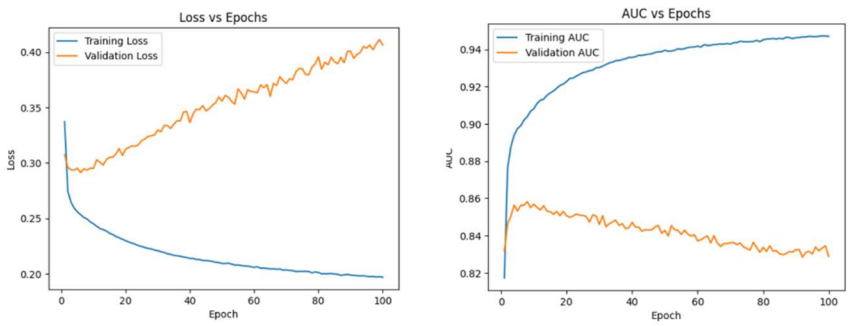


Figure 3.2: Tuned XGBoost feature importance

3.2 Neural Network and hyperparameter tuning

Neural network, 2 hidden layers (12, 8 neurons), relu activation, sigmoid output activation, optmiser adam, 100 epochs used. AUC (0.8505), higher than XGBoost, but lower than tuned XGBoost.



Loss, AUC curves - overfitting (validation lines diverge).

Figure 3.2: Neural Networks Loss, AUC curves

Accuracy reduced (figure 3.3), similar recall.

```
Test accuracy score for neural network:  0.8830806065442937

              precision    recall  f1-score   support

           0       0.92      0.94      0.93      4261
           1       0.62      0.55      0.59       751

    accuracy                           0.88      5012
   macro avg       0.77      0.75      0.76      5012
weighted avg       0.88      0.88      0.88      5012
```

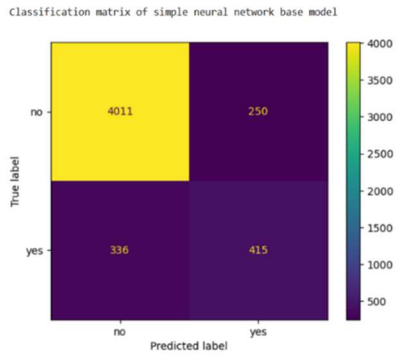
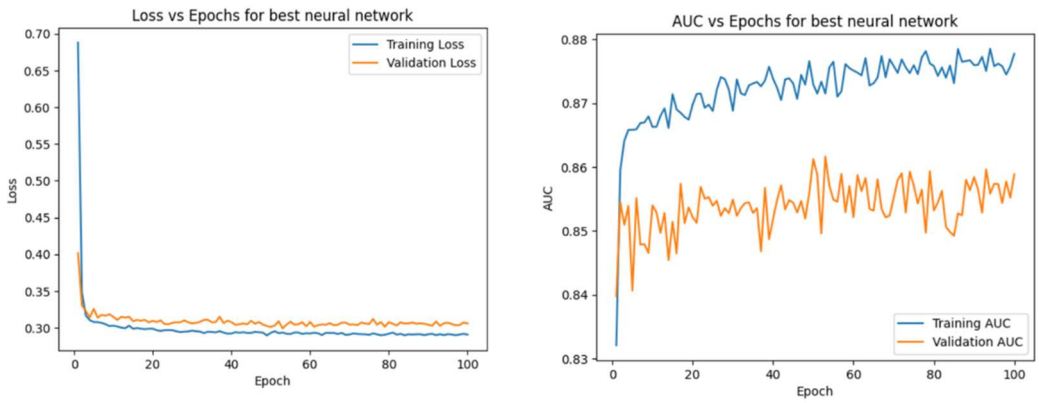


Figure 3.3: Neural networks

Tuning carried out 16, 32, 64 neurons (2nd layer halved), activation – sigmoid, tanh, optimiser – adam, rmsprop (figure 3.4).

	Neurons	Activation	Optimiser used	Test AUC
8	64	sigmoid	adam	0.875381
6	32	tanh	adam	0.874246
2	16	tanh	adam	0.873043
0	16	sigmoid	adam	0.872981
4	32	sigmoid	adam	0.872114
10	64	tanh	adam	0.867305
1	16	sigmoid	rmsprop	0.867065
9	64	sigmoid	rmsprop	0.866771
5	32	sigmoid	rmsprop	0.866684
7	32	tanh	rmsprop	0.844841
3	16	tanh	rmsprop	0.844399
11	64	tanh	rmsprop	0.840965

Figure 3.4: Best model, tuned Neural Networks



Validation curves don't diverge. AUC has divergence, signs of some overfitting.

Figure 3.4: Best tuned Neural Networks – Loss, AUC curves

Accuracy is higher, compared to base neural network, not as good as XGBoost (tuned and not-tuned), figure 3.5.

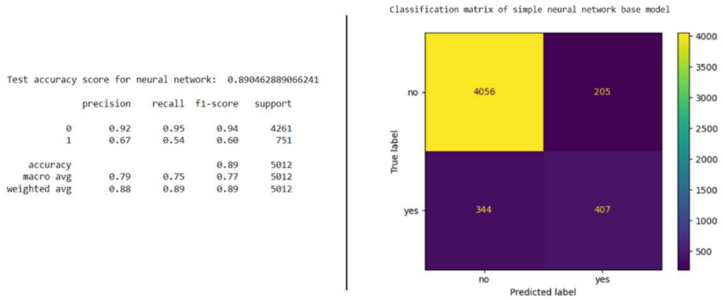


Figure 3.5: Neural Networks tuned

3.3 Analysis

XGBoost, tuned XGBoost, neural network and tuned neural network compared (figure 3.6). Recall and AUC picked as data imbalanced. The recall, accuracy and AUC plotted. AUC improves with tuned XGBoost, but recall goes lower. AUC gets better with tuned neural network, and recall goes lower.

Best AUC is XGBoost tuned, outperforming neural network a bit. However gives the least true positive values, most false negative. F1 scores are comparable.

	Test accuracy	Test AUC	Precision (0)	Precision (1)	Recall (0)	Recall (1)	F1 (0)	F1(1)	True Negative	True Positive	False Negative	False Positive
XGBoost	0.895450918	0.752592413	0.92	0.69	0.96	0.55	0.94	0.61	4076	412	339	185
XGBoost tuned	0.893655227	0.885334769	0.92	0.69	0.96	0.53	0.94	0.6	4081	398	353	180
Neural Network	0.883080607	0.8505	0.92	0.62	0.94	0.55	0.93	0.59	4011	415	336	250
Neural Network tuned	0.890462889	0.875381	0.92	0.67	0.95	0.54	0.94	0.6	4056	407	344	205

	Test accuracy	Test AUC	Recall (1)
XGBoost	0.895450918	0.752592413	0.55
XGBoost tuned	0.893655227	0.885334769	0.53
Neural Network	0.883080607	0.8505	0.55
Neural Network tuned	0.890462889	0.875381	0.54

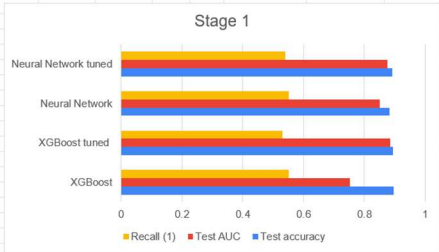


Figure 3.6: Comparison

4.0 Stage2

Stage2 data has 2 new features (figure 4.1) with 208 NaN values = 0.8% total records. As < 2%, its removed. Imbalance similar with small removal.

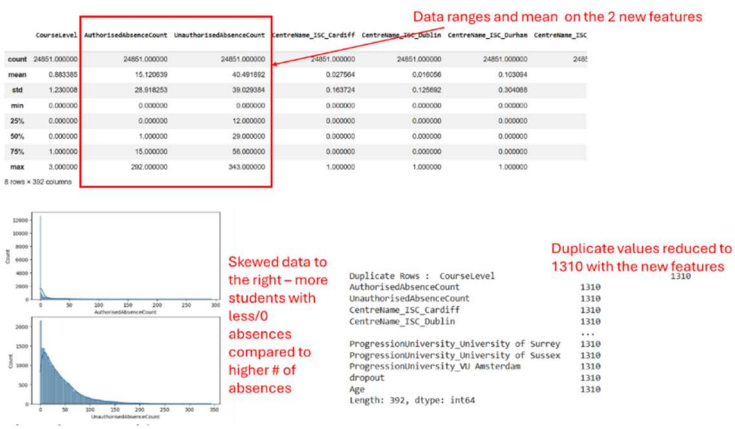


Figure 4.1: 2 new features data analysis

4.1 Base mode: XGBoost and Neural Network

XGBoost creates a higher accuracy, AUC, true positives and negatives, recall – in comparison to stage1 base model. New features added prediction value (figure 4.2).

```
XGBoost Stage 2 Model Accuracy:  0.9066586199959766
XGBoost Stage 2 Model AUC:  0.7723702627765112
              precision    recall  f1-score   support

           0       0.93      0.96      0.95      4249
           1       0.72      0.58      0.64       722

    accuracy                           0.91      4971
   macro avg       0.83      0.77      0.80      4971
weighted avg       0.90      0.91      0.90      4971
```

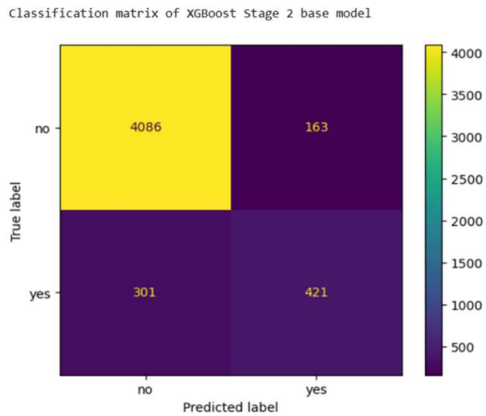
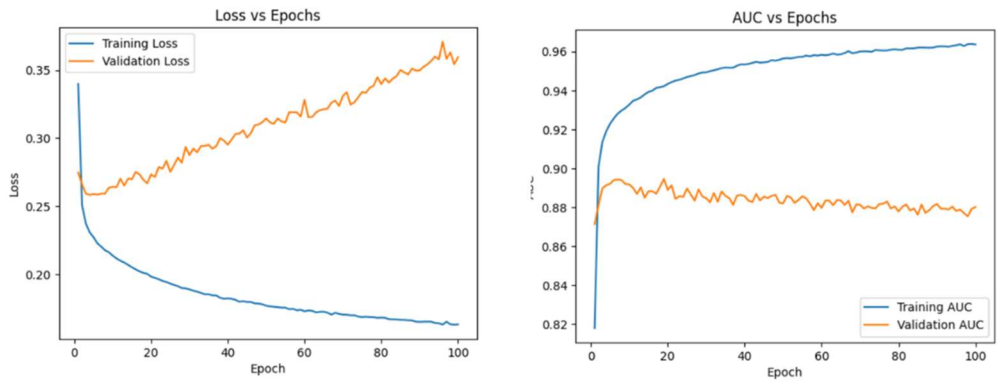


Figure 4.2: XGBoost



Neural network shows loss, AUC curves, similar to stage1 base model, has less loss, slightly higher AUC - overfitting.

Figure 4.3: Neural network Loss, AUC curves

Neural network (figure 4.4) has higher accuracy, AUC, true positives, true negatives, recall when compared to stage1 neural network base.

```
Test accuracy score for neural network:  0.8990142828404748

              precision    recall  f1-score   support

           0       0.93      0.95      0.94      4249
           1       0.67      0.60      0.63       722

    accuracy                           0.90      4971
   macro avg       0.80      0.77      0.79      4971
weighted avg       0.89      0.90      0.90      4971
```

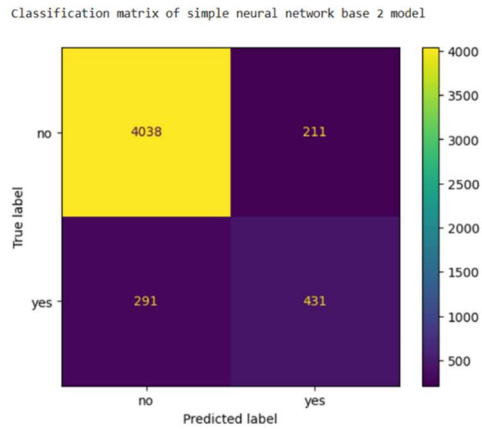


Figure 4.4: Neural network metrics

4.2 Hyperparameter tuning: XGBoost and Neural Network

XGBoost tuned produces (figure 4.5) higher AUC, true negatives and accuracy compared to stage1 tuned XGBoost and stage2 XGBoost base. Slightly lower true positives compared to base.

Best model

index	model	best_cv_auc	test_auc	test_accuracy	best_params
0	XGBoost Hyperparameter	0.9110266842758505	0.9111488836545538	0.9096772882719774	{'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200}

	precision	recall	f1-score	support
0	0.93	0.97	0.95	4249
1	0.74	0.58	0.65	722
accuracy			0.91	4971
macro avg	0.84	0.77	0.80	4971
weighted avg	0.90	0.91	0.91	4971

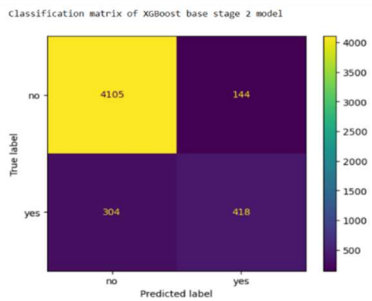
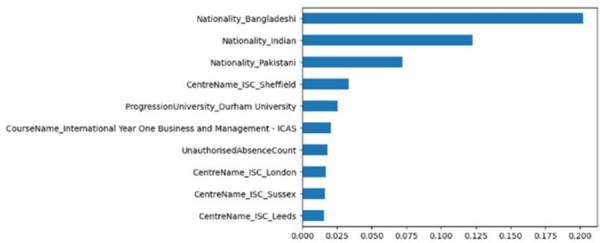


Figure 4.5: XGBoost tuned



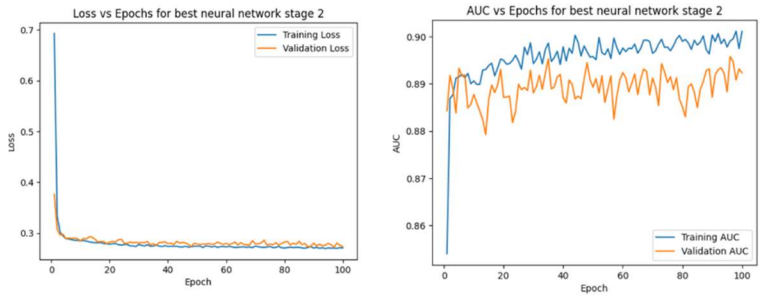
Change on Bangladeshi being higher than Indian, CourseName has come up. New feature - UnauthorisedAbsenceCount is in top 10 impacting prediction explaining the better performance.

Figure 4.6: XGBoost tuned feature importance

Neural networks tuning (figure 4.7).

	Neurons	Activation	Optimiser used	Test AUC
8	64	sigmoid	adam	0.886116
6	32	tanh	adam	0.885791
10	64	tanh	adam	0.885408
2	16	tanh	adam	0.884404
9	64	sigmoid	rmsprop	0.884237
1	16	sigmoid	rmsprop	0.882553
4	32	sigmoid	adam	0.880932
5	32	sigmoid	rmsprop	0.880797
0	16	sigmoid	adam	0.880407
7	32	tanh	rmsprop	0.874111
3	16	tanh	rmsprop	0.869969
11	64	tanh	rmsprop	0.869921

Figure 4.7: Neural network tuned best model



Less diverging graph– compared to stage1 base/tuned version, stage2 base.

Figure 4.8: Neural network tuned, Loss, AUC curves

```
Test accuracy score for neural network stage 2:  0.8988131160732247

              precision    recall  f1-score   support

           0       0.93      0.96      0.94      4249
           1       0.68      0.56      0.62       722

    accuracy                           0.90      4971
   macro avg       0.81      0.76      0.78      4971
weighted avg       0.89      0.90      0.89      4971
```

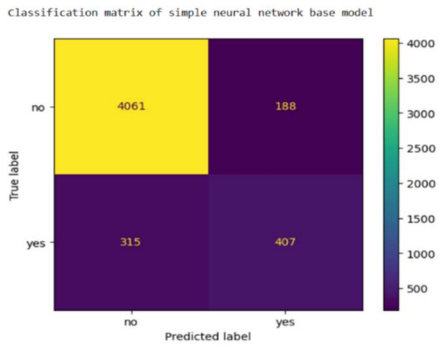


Figure 4.8: Neural network tuned metrics

Tuned neural network has better AUC, accuracy compared to stage1 (similar to base stage2). False negative (315) is less compared to stage1 (344), less of missed opportunity. Base performs better.

4.3 Analysis: Stage1 vs Stage2

Best accuracy, AUC is XGBoost tuned stage2 (highest performing model), Best recall is Neural Network base model stage2, 0.6 (figure4.9), but has overfitting. Tuned model for neural networks stage1 doesn't increase its accuracy/recall.

XGBoost tuned stage2 recall is 0.58 (close to 0.6). Stage2 performs better with new data.

		Test accuracy	Test AUC	Precision (0)	Precision (1)	Recall (0)	Recall (1)	F1 (0)	F1(1)	True Negative	True Positive	False Negative	False Positive
Stage 1	**XGBoost**	0.895450918	0.752592413	0.92	0.69	0.96	0.55	0.94	0.61	4076	412	339	185
	XGBoost tuned	0.893655227	0.885334769	0.92	0.69	0.96	0.53	0.94	0.6	4081	398	353	180
	Neural Network	0.883080607	0.8505	0.92	0.62	0.94	0.55	0.93	0.59	4011	415	336	250
	Neural Network tuned	0.890462889	0.875381	0.92	0.67	0.95	0.54	0.94	0.6	4056	407	344	205
Stage 2	**XGBoost**	0.90665862	0.772370263	0.93	0.72	0.96	0.58	0.95	0.64	4086	421	301	163
	XGBoost tuned	0.909877288	0.911148884	0.93	0.74	0.97	0.58	0.95	0.65	4105	418	304	144
	Neural Network	0.899014283	0.8606	0.93	0.67	0.95	0.6	0.94	0.63	4038	431	291	211
	Neural Network tuned	0.898813116	0.886116	0.93	0.68	0.96	0.56	0.94	0.62	4061	407	315	188

		Test accuracy	Test AUC	Recall (1)
Stage 1	**XGBoost**	0.895450918	0.752592413	0.55
	XGBoost tuned	0.893655227	0.885334769	0.53
	Neural Network	0.883080607	0.8505	0.55
	Neural Network tuned	0.890462889	0.875381	0.54
Stage 2	**XGBoost**	0.90665862	0.772370263	0.58
	XGBoost tuned	0.909877288	0.911148884	0.58
	Neural Network	0.899014283	0.8606	0.6
	Neural Network tuned	0.898813116	0.886116	0.56

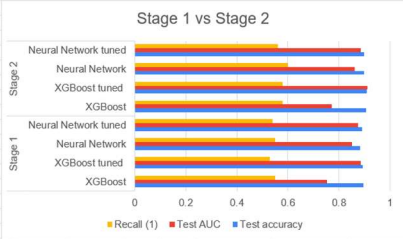


Figure 4.9: Stage1 vs stage2

5.0 Stage3

3 new features added, having 2231 null values = 8.9% of records. Its > 2%, these can't be removed, but imputed with 0. Could indicate good student with no failures or student dropped out (figure 5.1)

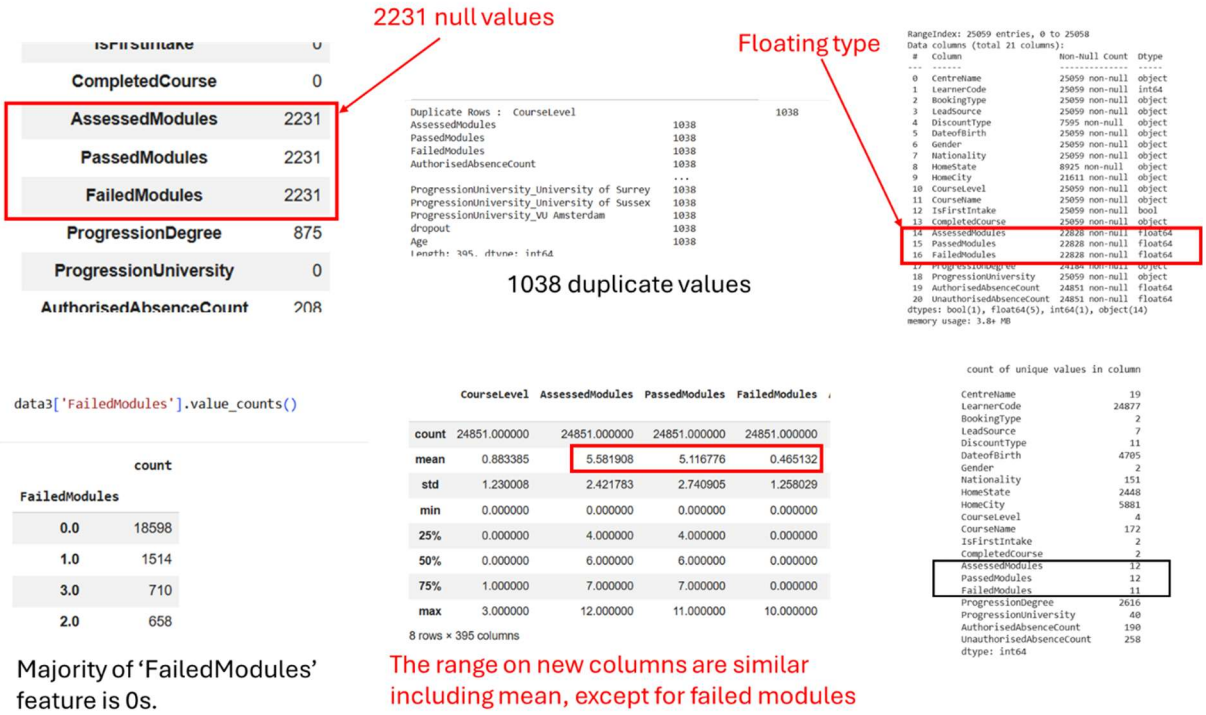
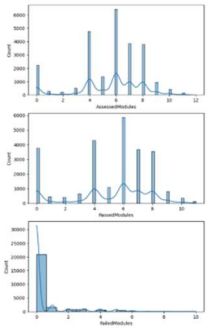


Figure 5.1: Data exploration



Distribution of the 3 new features. Last is skewed towards the 0s, as not many failed, with rest in middle ranges.

Figure 5.2: Data distribution

5.1 Base Model: XGBoost and Neural Network

Accuracy, AUC and recall is significantly better on both, with added features (figure 5.3). True positives/negatives go up. False negatives i.e. missed chances to intervene, gone down significantly with both base models.

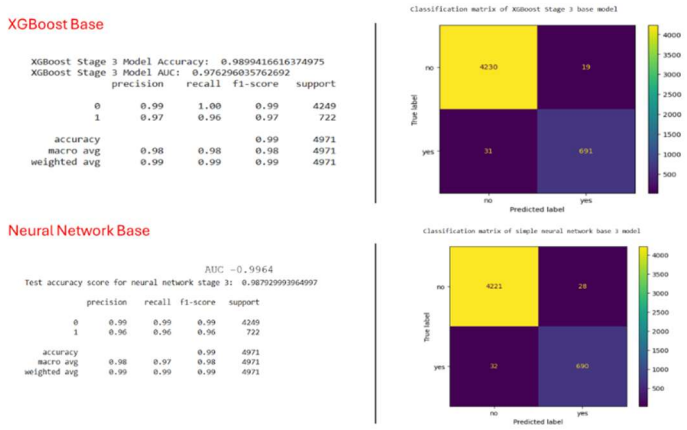
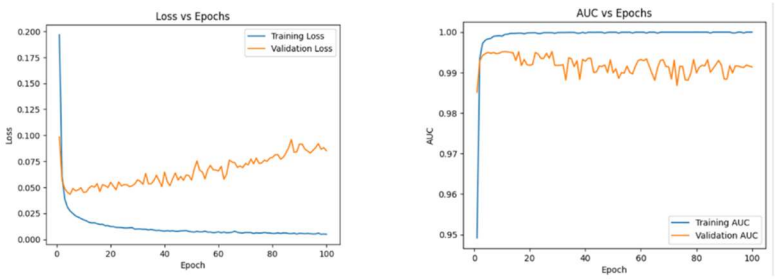


Figure 5.3: XGBoost and Neural Network metrics



Loss, AUC curves for neural network shows reduction in loss, and AUC rise. Split exists, i.e. overfitting in smaller quantities.

Figure 5.4: Neural network loss, AUC curves

5.2 Hyperparameter tuning: XGBoost and Neural Network

XGBoost tuned performs better in accuracy, auc and recall (recall is same to stage3 base).

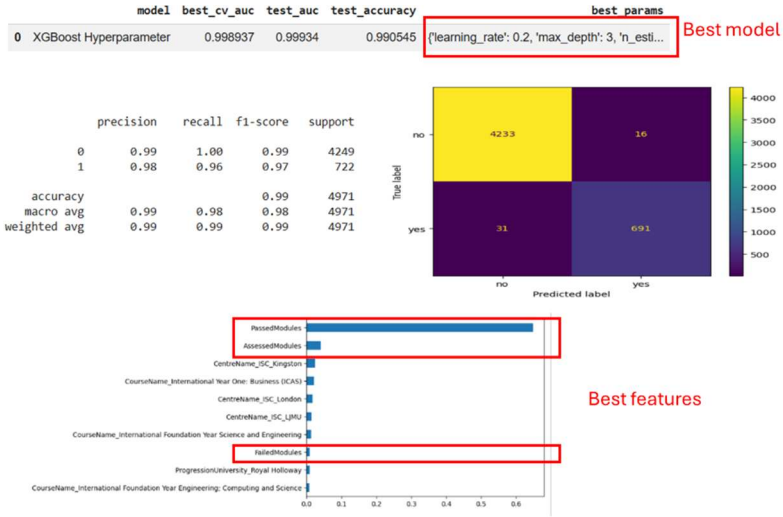


Figure 5.5: XGBoost tuned - best model, feature and metrics

Neural networks tuned perform better (figure 5.6) in comparison to stage2 neural networks tuned. Recall significantly increases. Missed predictions drops significantly. This doesn't outperform XGBoost tuned. Similar but XGBoost is better.

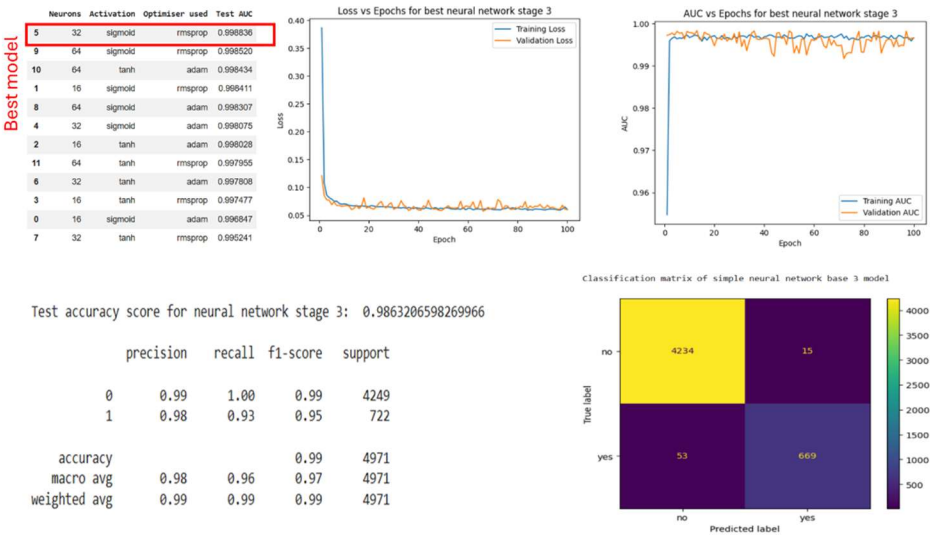


Figure 5.6: Neural network best model, loss and auc curves

5.3 Analysis: Stage2 vs Stage3 dataset results

Comparing Stage2 vs Stage3 results – Highest accuracy, AUC and Recall is with XGBoost tuned for Stage3. Neural network doesn't come up to same level as XGBoost. The 'false negatives' is lowest for XGBoost tuned. XGBoost caters for the added features better than neural network.

		Test accuracy	Test AUC	Precision (0)	Precision (1)	Recall (0)	Recall (1)	F1 (0)	F1(1)	True Negative	True Positive	False Negative	False Positive
Stage 1	XGBoost	0.895450918	0.752592413	0.92	0.69	0.96	0.55	0.94	0.61	4076	412	339	185
	XGBoost tuned	0.893655227	0.885334769	0.92	0.69	0.96	0.53	0.94	0.6	4081	398	353	180
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Stage 2	XGBoost	0.90665862	0.772370263	0.93	0.72	0.96	0.58	0.95	0.64	4086	421	301	163
	XGBoost tuned	0.909877288	0.911148884	0.93	0.74	0.97	0.58	0.95	0.65	4105	418	304	144
	Neural Network	0.899014283	0.8606	0.93	0.67	0.95	0.6	0.94	0.63	4038	431	291	211
	Neural Network tuned	0.898813116	0.886116	0.93	0.68	0.96	0.56	0.94	0.62	4061	407	315	188
Stage 3	XGBoost	0.989941662	0.976296036	0.99	0.97	1	0.96	0.99	0.97	4230	691	31	19
	XGBoost tuned	0.990545	0.99934	0.99	0.98	1	0.96	0.99	0.97	4233	691	31	16
	Neural Network	0.987929994	0.9964	0.99	0.96	0.99	0.96	0.99	0.96	4221	690	32	28
	Neural Network tuned	0.98632066	0.998836	0.99	0.98	1	0.93	0.99	0.95	4234	669	53	15

		Test accuracy	Test AUC	Recall (1)
Stage 2	XGBoost	0.90665862	0.772370263	0.58
	XGBoost tuned	0.909877288	0.911148884	0.58
	Neural Network	0.899014283	0.8606	0.6
	Neural Network tuned	0.898813116	0.886116	0.56
Stage 3	XGBoost	0.989941662	0.976296036	0.96
	XGBoost tuned	0.990545	0.99934	0.96
	Neural Network	0.987929994	0.9964	0.96
	Neural Network tuned	0.98632066	0.998836	0.93

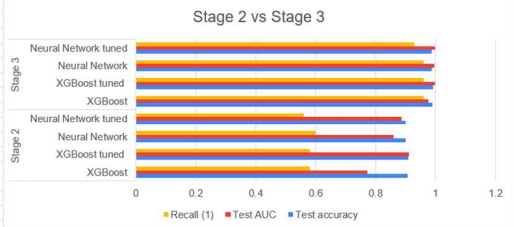

Figure 5.7: Stage2 vs Stage3

6.0 Conclusion and recommendations

The tuned model for XGBoost on Stage3 data outperformed the earlier stages. There is a significant improvement through the stages due to additional features.

Using the XGBoost tuned model with stage3 data helps predict student dropout rate accurately, with 31 students who might have been classed as 'non dropout' but are actual dropout. Keeping this number low helps with retention and save money for the institution. High accuracy, AUC and recall performance metrics reflects this.

Other metrics to consider for prediction for improvement. Modular code to be used for future.

References

Data, 2026. *Data Science programme at the University of Cambridge*: https://fourthrev.instructure.com/courses/991/assignments/3264?module_item_id=76829 [Accessed: 14 March 2026]

Data Science, 2026. *Data Science* programme at the University of Cambridge [Accessed: 14 March 2026]